Solution Tutorial VI – Take-Off and Landing

Task 1

For take-off, the flap lever is set to "2". Determine the achievable lift coefficients due to trailing edge flaps ($\Delta C_{L,TE}$) and due to leading edge slats ($\Delta C_{L,LE}$). Use these results to calculate the maximum lift coefficient in take-off configuration $C_{L,max,TO}$.

- Position "2":

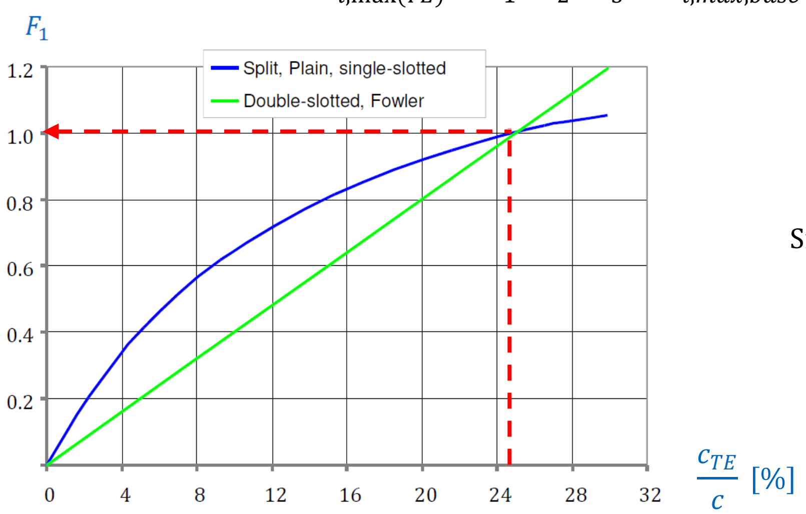
$\delta_{LE} = 22°$

$\delta_{TE} = 15°$

- <u>Trailing edge:</u>

1) Airfoil:

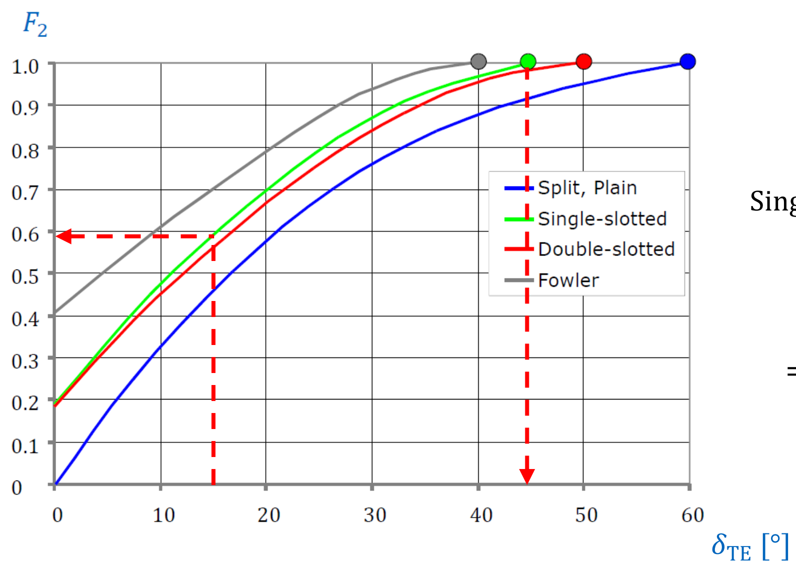
$$\Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base}$$



$$\frac{c_{TE}}{c} = \frac{1.05}{4.1935} = 25.0\ \%$$

Single slotted $\rightarrow$ blue curve

$$\Longrightarrow F_1 = 1.0$$

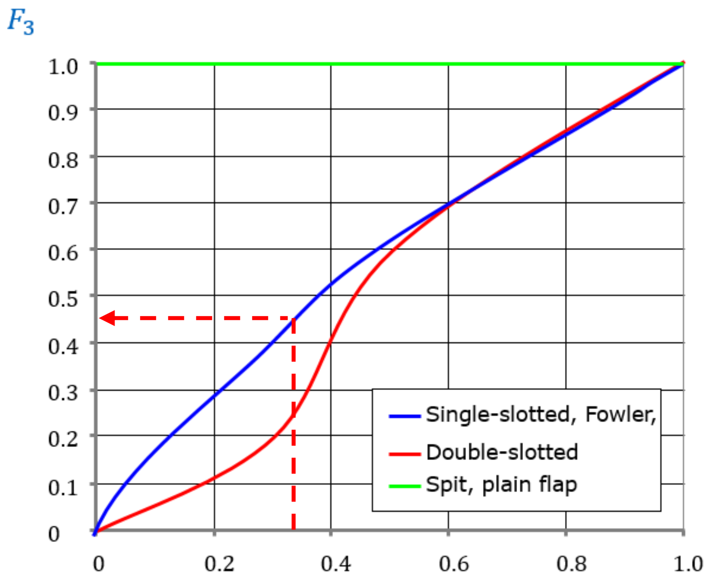


$$\delta_{TE} = 15°$$

Single slotted $\rightarrow$ green curve

$$\Longrightarrow F_2 = 0.58$$

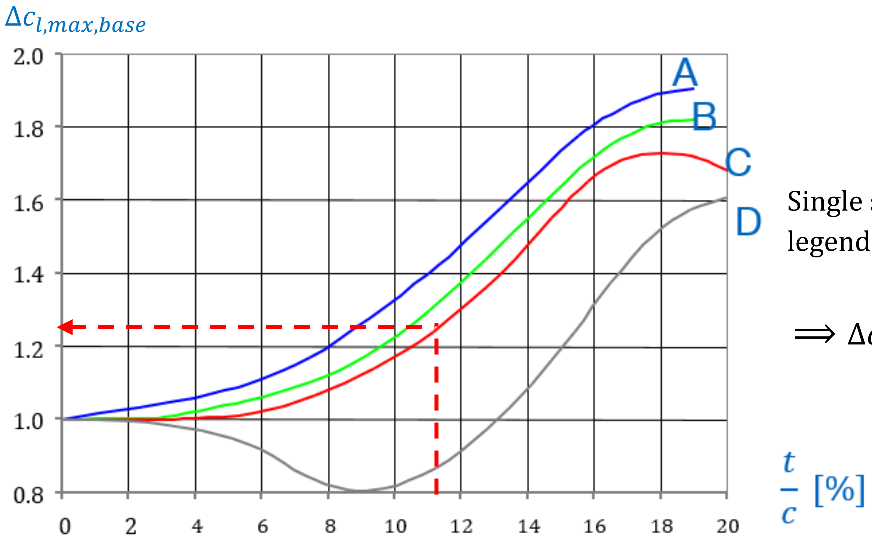
$$\Longrightarrow \delta_{ref} = 45° \ @$$



$$\frac{\delta_{TE}}{\delta_{ref}} = \frac{15°}{45°} = 0.333$$

Single slotted $\rightarrow$ blue curve

$$\Longrightarrow F_3 = 0.46$$



$$\frac{t}{c} = 11.75\ \%$$

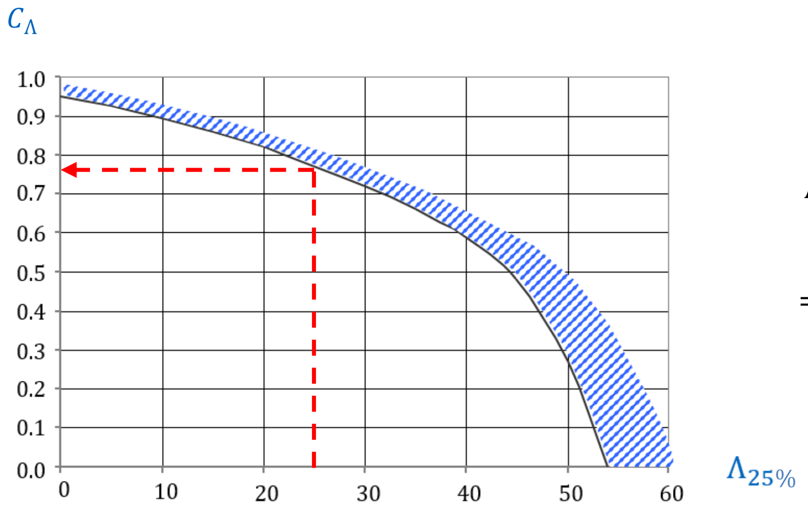
Single slotted $\rightarrow$ curve C (see legend in problem formulation)

$$\Longrightarrow \Delta c_{l,max,base} = 1.24$$

$$\Longrightarrow \Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base} = 1 \cdot 0.58 \cdot 0.46 \cdot 1.24 = 0.331$$

2) Wing:

$$\Delta C_{L,max(TE)} = \Delta c_{l,max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_{\Lambda}$$



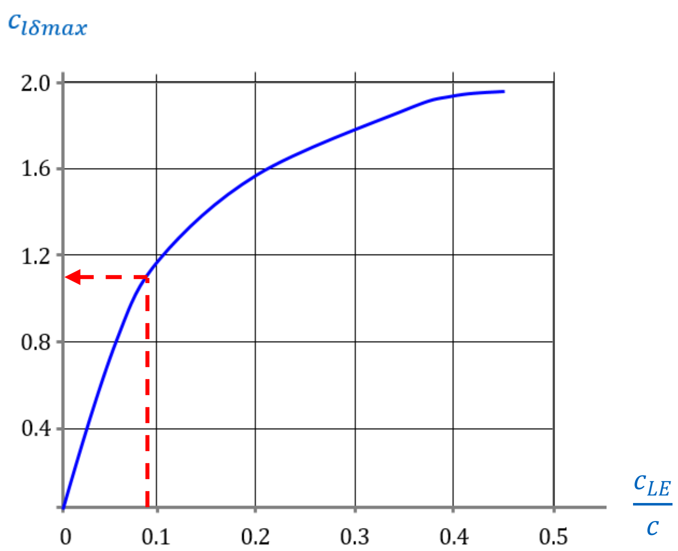
$$\Lambda_{25\%} = 25.0°$$

$$\Longrightarrow C_\Lambda = 0.78$$

$$\Longrightarrow \Delta C_{L,\max(TE)} = \Delta c_{l,\max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_\Lambda = 0.331 \cdot 74.7\% \cdot 0.78 = 0.193$$

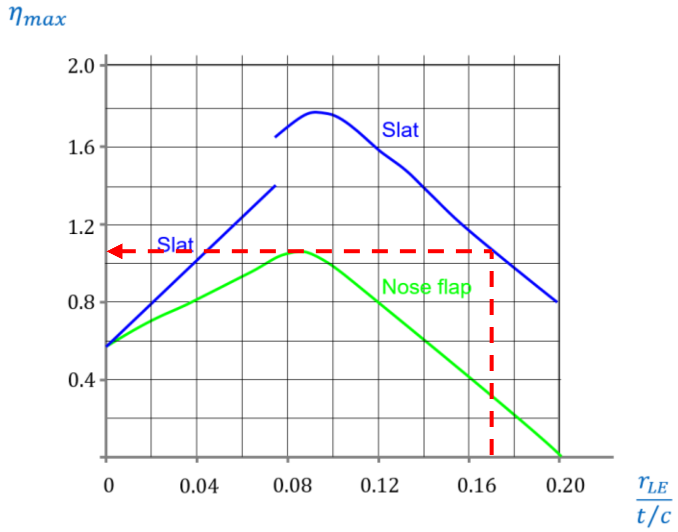
- <u>Leading edge:</u>
  1) Airfoil:

$$\Delta c_{l,\max(LE)} = c_{l\delta max} \cdot \eta_{max} \cdot \delta_{LE} \cdot \left(\frac{c + c_{LE}}{c}\right) \cdot \pi/180$$



$$\frac{c_{LE}}{c} = \frac{0.4}{4.1935} = 0.0954$$

$$\Longrightarrow c_{l\delta max} = 1.10$$



$$\frac{r_{LE}}{t/c} = \frac{0.02}{11.75\,\%} = 0.170$$

Slat $\rightarrow$ blue curve

$$\Longrightarrow \eta_{max} = 1.07$$

$$\Longrightarrow \Delta c_{l,\max(LE)} = c_{L\delta max} \cdot \eta_{max} \cdot \delta_{LE} \cdot \left(\frac{c + c_{LE}}{c}\right) \cdot \pi/180$$

$$= 1.1 \cdot 1.07 \cdot 22 \cdot \left(\frac{4.1935 + 0.4}{4.1935}\right) \cdot \pi/180 = 0.495$$

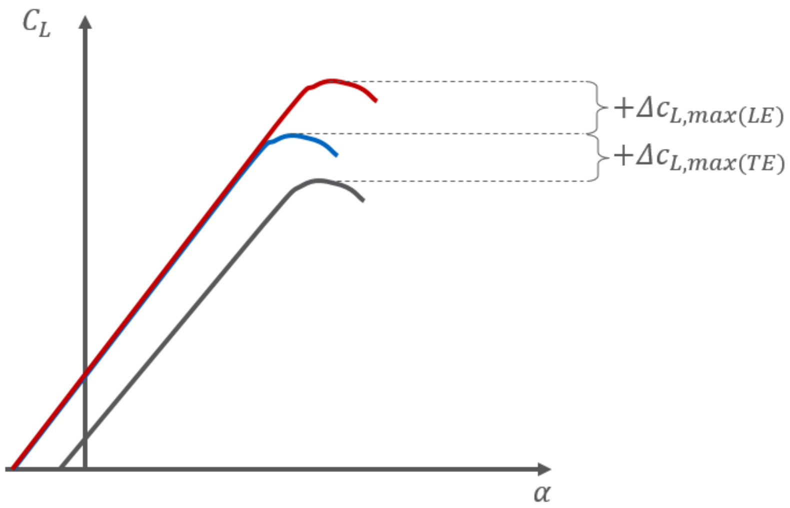
2) Wing:

$$\Delta C_{L,max(LE)} = \Delta c_{l,\max(LE)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos^2(\Lambda_{25\%}) = 0.495 \cdot 72.8\% \cdot 0.9066^2 = 0.296$$

- <u>Total:</u>

$$C_{L,max,TO} = C_{L,max,clean} + \Delta C_{L,max(TE)} + \Delta C_{L,max(LE)}$$

$$\Longrightarrow C_{L,max,TO} = 1.5 + 0.193 + 0.296 = 1.99$$

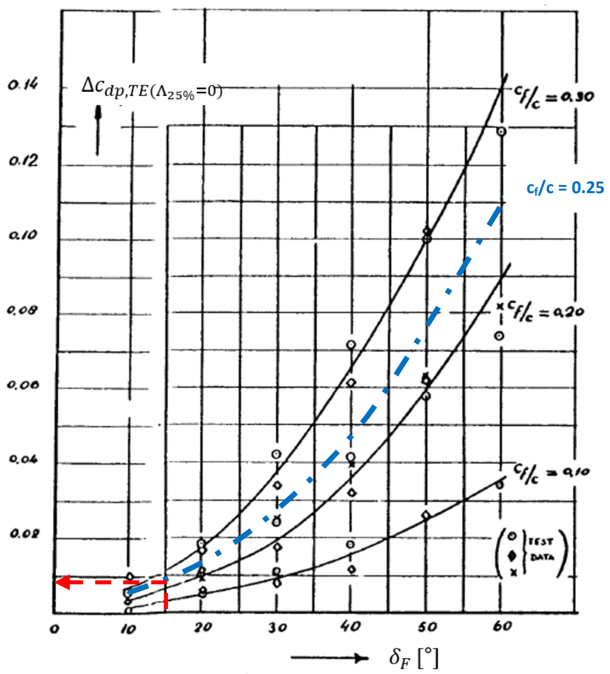


Task 2

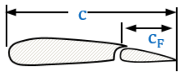
Also calculate the drag increments $\Delta C_{D,TE}$ and $\Delta C_{D,LE}$ in this case using the Roskam method. Use these results to calculate the total drag coefficient increment due to high-lift devices in take-off configuration $\Delta C_{D,HLD,TO}$.

- <u>Trailing edge drag increment:</u>

$$\Delta C_{D,TE} = \Delta c_{dp,TE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$



Single slotted flaps



Single slotted flaps at trailing edge
→ choose diagram accordingly

$$\frac{c_f}{c} = \frac{c_{TE}}{c} = \frac{1.05}{4.1935}$$

$$\delta_F = \delta_{TE} = 15°$$

$$\Longrightarrow \Delta c_{dp,TE(\Lambda_{25\%}=0)} = 0.008$$

$$\Longrightarrow \Delta C_{D,TE} = \Delta c_{dp,TE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot \cos(\Lambda_{25\%}) = 0.008 \cdot 74.7\% \cdot 0.9066 = 0.00542$$

- <u>Leading edge drag increment:</u>

$$\Delta C_{D,LE} = \Delta c_{dp,LE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

Leading edge system consists of slats, thus:

$$\Longrightarrow \Delta C_{D,LE} = C_{D0,W} \cdot \frac{c + \Delta c_{LE\,(TO)}}{c} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

$C_{D0,W}$ calculation using given parasite drag area f_W:

$$C_{D0,W} = \frac{f_W}{S_{wet,W}} = \frac{0.77}{2 * 122.4} = 0.00315$$

$$\Longrightarrow \Delta C_{D,LE} = 0.00315 \cdot \frac{4.1935 + 0.05}{4.1935} \cdot 72.8\% \cdot 0.9066 = 0.00210$$

- HLD drag increment:
  $$\Delta C_{D,HLD,TO} = \Delta C_{D,TE} + \Delta C_{D,LE} = 0.00542 + 0.00210 = 0.00752$$

Task 3

Now calculate the take-off distance of the Airbus A320 at maximum take-off mass from an airport located at mean sea level. Break the calculation down to:

Acceleration distance s_{acc} (with $\bar{T} \approx T_0 \cdot \frac{3}{4} \cdot \frac{5+BPR}{4+BPR}$)
Rotation distance s_{rot}
Transition distance $s_{T,min}$

- Acceleration distance:
  Equation to be used:

$$\mathrm{s_{acc}} = \frac{C^2 \cdot \frac{w_{TO}}{S_W}}{\rho \cdot g \cdot [C_{L,max,TO} \cdot \left(\frac{\bar{T}}{w_{TO}} - \mu_f\right) - \frac{C^2}{2} \cdot (\mathrm{C}_{D0|clean} + \Delta C_{D,HLD,TO} + \Delta C_{D,LG} + \frac{{C_{L_{TO}}}^2}{3 \cdot \pi \cdot AR \cdot e_{TO}} - \mu_f \cdot C_{L,TO})]}$$

  Coefficients of equation:
  - $C = 1.15$ (safety factor)
  - $w_{TO} = m_{TO} \cdot g = 73500 \cdot 9.81 = 721000\ N$ (take-off weight)
  - $S_W = 122.4\ m^2$ (wing reference area)
  - $g = 9.81\ m/s^2$ (gravitational acceleration)
  - $\rho = 1.23\ \ kg/m^3$ (air density)
  - $C_{L,max,TO} = 1.99$ (max. lift coefficient, Task 1)
  - $\bar{T} \approx T_0 \cdot \frac{3}{4} \cdot \frac{5+BPR}{4+BPR} = 2 \cdot 118000\ \cdot \frac{3}{4} \cdot \frac{5+5.5}{4+5.5} = 196000\ N$ (average thrust at take-off)
  - $\mu_f = 0.02$ (rolling friction coefficient)
  - $C_{D0|clean} = 0.018$ (zero-lift drag coefficient)
  - $\Delta C_{D,HLD,TO} = 0.00752$ (drag coefficient increment, Task 2)
  - $\Delta C_{D,LG} = 1/4 \cdot 0.018 = 0.0045$ (landing gear drag coefficient)
  - $C_{L,TO} = \frac{C_{L,max,TO}}{C^2} = \frac{1.99}{1.15^2} = 1.50$ (take-off lift coefficient)
  - $\mathrm{AR} = 9.39$ (aspect ratio)
  - $e_{TO} = 0.80$ (Oswald factor at take-off)

  Therefore:

$$\mathrm{s_{acc}} = \frac{1.15^{\ 2} \cdot \frac{721000}{122.4}}{1.23 \cdot 9.81 \cdot \left[1.99 \cdot \left(\frac{196000}{721000} - 0.02\right) - \frac{1.15^{\ 2}}{2} \cdot \left(0.018 + 0.00752 + 0.0045 + \frac{1.50^2}{3 \cdot \pi \cdot 9.39 \cdot 0.80} - 0.02 \cdot 1.50\right)\right]}$$

$$\Longrightarrow \mathrm{s_{acc}} = 1340\ m$$

- Rotation distance:
  Equation to be used:

$$s_{rot} = t_{rot} \cdot V_{TO}$$

Coefficients of equation:
- $t_{rot} = \Delta t = 3\ sec$ (rotation duration)
- $V_{TO} = \sqrt{\frac{2 \cdot C^2 \cdot W_{TO}/S_W}{\rho \cdot C_{L,max,TO}}} = \sqrt{\frac{2 \cdot 1.15^2 \cdot 721000/122.4}{1.23 \cdot 1.99}} = 79.8\ m/s$ (take-off airspeed)

Therefore:

$$\Longrightarrow s_{rot} = 3 \cdot 79.8 = 239\ m$$

- <u>Transition distance:</u>
  Equation to be used:

$$s_{T,min} = \sqrt{\frac{2 \cdot h \cdot V_{TO}^2}{g \cdot (n_z - 1)}}$$

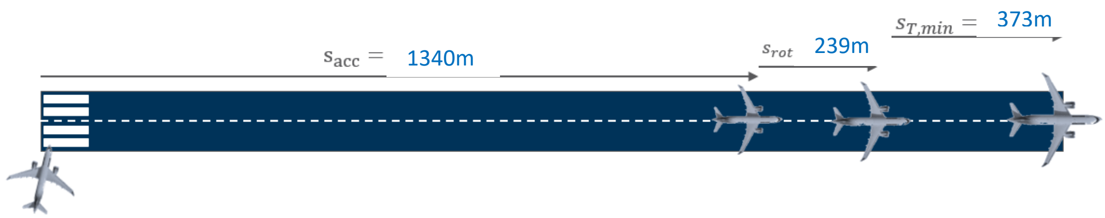
Coefficients of equation:
- $h = 35\ ft = 10.7\ m$ (transition height)
- $n_z = 1.1$ (take-off load factor)
- $g = 9.81\ m/s^2$ (gravitational acceleration)

Therefore:

$$\Longrightarrow s_{T,min} = \sqrt{\frac{2 \cdot 10.7 \cdot 79.8^2}{9.81 \cdot (1.1 - 1)}} = 373\ m$$

- <u>Total take-off distance:</u>

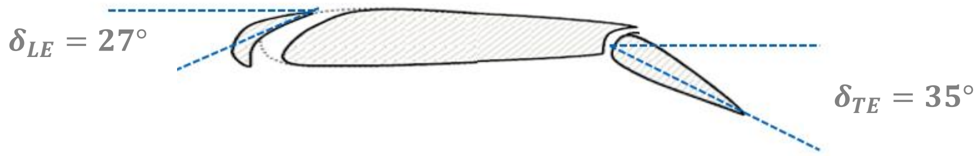
$$s_{TO} = s_{acc} + s_{rot} + s_{T,min} = 1340 + 239 + 373 = 1950\ m$$



Task 4

For landing, the flap lever is set to "full". Determine the achievable lift coefficients due to trailing edge flaps ($\Delta C_{L,TE}$) and due to leading edge slats ($\Delta C_{L,LE}$). Use these results to calculate the maximum lift coefficient in landing configuration $C_{L,max,L}$.

- Position “full”:



- Trailing edge:
  1) Airfoil:

$$\Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base}$$

Using the same diagrams as in task 1:

$$\frac{c_{TE}}{c} = \frac{1.05}{4.1935} = 25.0\ \%\ ;\ \text{single slotted} \rightarrow \text{blue curve} \implies F_1 = 1.0$$

$$\delta_{TE} = 35°\ \text{single slotted} \rightarrow \text{green curve} \implies F_2 = 0.94\quad \& \quad \implies \delta_{ref} = 45°$$

$$\frac{\delta_{TE}}{\delta_{ref}} = \frac{35°}{45°} = 0.77;\ \text{single slotted} \rightarrow \text{blue curve} \implies F_3 = 0.81$$

$$\frac{t}{c} = 11.75\ \%\ ;\ \text{single slotted} \rightarrow \text{curve C} \implies \Delta c_{l,max,base} = 1.24$$

Therefore:

$$\implies \Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base} = 1 \cdot 0.94 \cdot 0.81 \cdot 1.24 = 0.944$$

  2) Wing:

$$\Delta C_{L,\max(TE)} = \Delta c_{l,\max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_{\Lambda}$$

Using the same diagram as in task 1:

$$\Lambda_{25\%} = 25.0° \quad \implies C_{\Lambda} = 0.78$$

Therefore:

$$\implies \Delta C_{L,max(TE)} = \Delta c_{l,max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_{\Lambda} = 0.944 \cdot 74.7\% \cdot 0.78 = 0.550$$

- Leading edge:
  1) Airfoil:

$$\Delta c_{l,\max(LE)} = c_{l\delta max} \cdot \eta_{max} \cdot \delta_{LE} \cdot \left(\frac{c + c_{LE}}{c}\right) \cdot \frac{\pi}{180}$$

Using the same diagrams as in task 1:

$$\frac{c_{LE}}{c} = \frac{0.4}{4.1935} = 0.095 \implies c_{l\delta max} = 1.1$$

$$\frac{r_{LE}}{t/c} = \frac{0.02}{11.75\%} = 0.17\ ;\ \text{slat} \rightarrow \text{blue curve} \implies \eta_{max} = 1.07$$

Therefore:

$$\Longrightarrow \Delta c_{l,\max(LE)} = c_{L\delta max} \cdot \eta_{max} \cdot \delta_{LE} \cdot \left(\frac{c + c_{LE}}{c}\right) \cdot \pi/180$$

$$= 1.1 \cdot 1.07 \cdot 27 \cdot \left(\frac{4.1935 + 0.4}{4.1935}\right) \cdot \pi/180 = 0.608$$

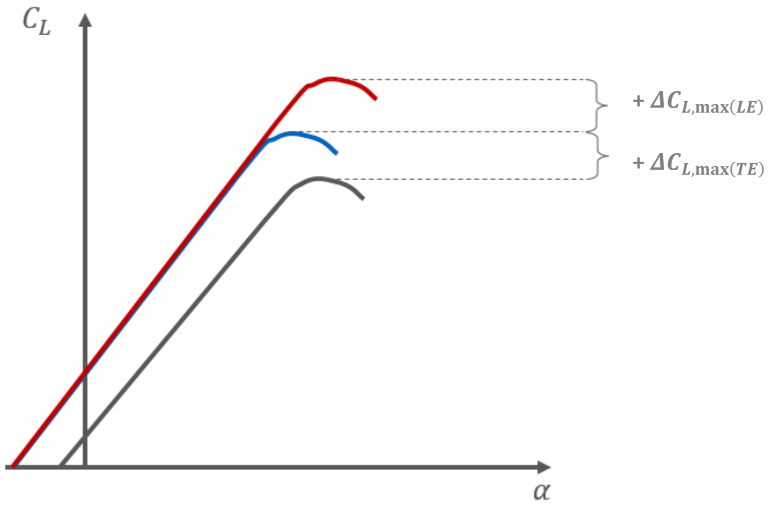
2) Wing:

$$\Delta C_{L,max(LE)} = \Delta c_{l,\max(LE)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos^2(\Lambda_{25\%}) = 0.608 \cdot 72.8\% \cdot 0.9066^2 = 0.364$$

- Total:

$$C_{L,max,L} = C_{L,max,clean} + \Delta C_{L,\max(TE)} + \Delta C_{L,\max(LE)}$$

$$\Longrightarrow C_{L,max,L} = 1.5 + 0.550 + 0.364 = 2.41$$



Task 5

Also calculate the drag increments $\Delta C_{D,TE}$ and $\Delta C_{D,LE}$ in this case using the Roskam method. Use these results to calculate the total drag coefficient increment due to high-lift devices in landing configuration $\Delta C_{D,HLD,L}$.

- Trailing edge drag increase:

$$\Delta C_{D,TE} = \Delta c_{dp,TE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

Using the same diagram as in task 2:

$$\delta_F = \delta_{TE} = 35° \Longrightarrow \Delta c_{dp,TE(\Lambda_{25\%}=0)} = 0.036$$

Therefore:

$$\Longrightarrow \Delta C_{D,TE} = 0.036 \cdot 74.7\% \cdot 0.9066 = 0.0244$$

- Leading edge drag increase:

$$\Delta C_{D,LE} = \Delta c_{dp,LE(\Lambda_{25\%}=0)} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

Leading edge system consists of slats, thus:

$$\Longrightarrow \Delta C_{D,LE} = C_{D0,W} \cdot \frac{c + \Delta c_{LE\,(L)}}{c} \cdot \frac{S_{W,LE}}{S_{ref}} \cdot \cos(\Lambda_{25\%})$$

$C_{D0,W}$ calculation using given parasite drag area f_W:

$$C_{D0,W} = \frac{f_W}{2 * S_{ref}} = \frac{0.77}{2 * 122.4} = 0.00315$$

$$\Longrightarrow \Delta C_{D,LE} = 0.00315 \cdot \frac{4.1935 + 0.08}{4.1935} \cdot 72.8\% \cdot 0.9066 = 0.00212$$

- HLD drag increase:

$$\Delta C_{D,\mathrm{HLD},L} = \Delta C_{D,TE} + \Delta C_{D,LE} = 0.0244 + 0.00212 = 0.0265$$

Task 6

Finally, calculate the landing distance of the Airbus A320 at maximum landing mass from an airport located at mean sea level. Break the calculation down to:

Approach distance s_{L1}
Rotation distance $s_{L2,1}$
Deceleration distance s_{dec}

- Approach distance

Equation to be used:

$$s_{L1} = \left(\frac{\bar{C}_L}{\bar{C}_D}\right) \cdot \left[h + \frac{1}{2 \cdot g} \cdot \left(V_{ap}^2 - V_{TD}^2\right)\right]$$

Coefficients of equation:

- $$\bar{C}_L = \frac{\bar{C}_{L,TD} + \bar{C}_{L,ap}}{2} = \frac{\frac{2\,W}{\rho\,S\,V_{TD}^2} + \frac{2\,W}{\rho\,S\,V_{ap}^2}}{2} = \frac{\frac{2\,W}{\rho\,S\,1.15^2\,V_S^2} + \frac{2\,W}{\rho\,S\,1.3^2\,V_S^2}}{2} = \frac{\frac{C_{L,max}}{1.15^2} + \frac{C_{L,max}}{1.3^2}}{2}$$

  $$= \frac{2.41/1.15^2 + 2.41/1.3^2}{2} = 1.62 \text{ (average lift coefficient)}$$

- $$\bar{C}_D = C_{D0|clean} + \Delta C_{D,HLD,L} + \Delta C_{D,LG} + k \cdot \bar{C}_L^2$$

  $$= C_{D0|clean} + \Delta C_{D,HLD,L} + \Delta C_{D,LG} + \frac{1}{\pi \cdot AR \cdot e_L} \cdot \bar{C}_L^2$$

  $$= 0.018 + 0.0265 + 0.0045 + \frac{1}{\pi \cdot 9.39 \cdot 0.75} \cdot 1.62^2 = 0.168$$

  (average drag coefficient)

- $h = 50\,ft = 15.2\,m$ (transition height)

- $g = 9.81\,m/s^2$ (gravitational acceleration)

- $V_S = \sqrt{\frac{2}{\rho \cdot C_{L,max,L}} \cdot \frac{w_L}{S_W}} = \sqrt{\frac{2}{1.23 \cdot 2.41} \cdot \frac{64500 \cdot 9.81}{122.4}} = 59.1\ m/s$ (minimum airspeed)
- $V_{ap} = 1.3 \cdot V_S = 76.8\ m/s$ (approach airspeed)
- $V_{TD} = 1.15 \cdot V_S = 68.0\ m/s$ (touch-down airspeed)

Therefore:

$$s_{L1} = \left(\frac{1.62}{0.168}\right) \cdot \left[15.2 + \frac{1}{2 \cdot 9.81} \cdot (76.8^2 - 68.0^2)\right] = 773\ m$$

- <u>Rotation distance</u>

Equation to be used:

$$s_{L2,1} = \Delta t \cdot V_{TD}$$

Coefficients of equation:

- $\Delta t = 3\ sec$ (de-rotation duration)

Therefore:

$$s_{L2,1} = 3 \cdot 68.0 = 204\ m$$

- <u>Deceleration distance</u>

Equation to be used:

$$s_{dec} = \frac{1.15^2}{\bar{a}} \cdot \frac{1}{C_{L,max,L}} \cdot \frac{1}{\rho} \cdot \frac{w_L}{S_W}$$

Coefficients of equation:

- $a = 3.9$ m/s² (braking deceleration), Engine reverse thrust availability: $\eta_{TR} = 0.45$
- $\bar{a} = a + \frac{T_0 \cdot \eta_{TR}}{w_L} \cdot g = 3.9 + \frac{2 \cdot 118000 \cdot 0.45}{64500 \cdot 9.81} \cdot 9.81 = 5.55\ m/s^2$

(deceleration with thrust reversal)

Therefore:

$$s_{dec} = \frac{1.15^2}{5.55} \cdot \frac{1}{2.41} \cdot \frac{1}{1.23} \cdot \frac{64500 \cdot 9.81}{122.4} = 416\ m$$

- <u>Total landing distance:</u>

$$s_L = s_{L1} + s_{L2,1} + s_{dec} = 773 + 204 + 416 = 1390\ m$$